Singular Perturbations, Moments and Descriptor Systems: Linear

Yoonjae Hwang

30 October 2025

1 Linear Systems - Thinking about explicit solutions and moments

Consider the following linear time-invariant, SISO system:

$$\begin{aligned} \dot{x} &= Ax + Bu, \\ y &= Cx, \end{aligned} \tag{1}$$

$$\begin{aligned} \dot{w} &= Sw, \\ u &= Lw, \end{aligned} \tag{2}$$

Given initial conditions $x(0) = x_0$ and $w(0) = w_0$, the solution to the system is given by

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}BLe^{S\tau}w_0 d\tau. \tag{3}$$

The output of the system is given by

$$y(t) = Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}BLe^{S\tau}w_0 d\tau. \tag{4}$$

If $\sigma(A) \cup \sigma(S) = \emptyset$, then there exists a unique solution Π to the Sylvester equation

$$A\Pi + BL = \Pi S. \tag{5}$$

Using this, we can substitute an expression for BL into the integral term for $x(t)$:

$$\begin{aligned} x(t) &= e^{At}x_0 + \int_0^t e^{A(t-\tau)}(\Pi S - A\Pi)e^{S\tau}w_0 d\tau \\ &= e^{At}x_0 + \int_0^t e^{A(t-\tau)}\Pi S e^{S\tau}w_0 d\tau - \int_0^t e^{A(t-\tau)}A\Pi e^{S\tau}w_0 d\tau \\ &= e^{At}x_0 + \left(\Pi e^{St} - e^{At}\Pi\right)w_0. \end{aligned} \tag{6}$$

Forward Moments are defined to be

$$M_g = C\Pi \tag{7}$$

2 Problem Setup

Consider the following linear singularly perturbed system:

$$\begin{aligned} \dot{x}_1(t) &= A_{11}x_1(t) + A_{12}x_2(t) + B_1u(t), \\ \varepsilon\dot{x}_2(t) &= A_{21}x_1(t) + A_{22}x_2(t) + B_2u(t), \end{aligned} \tag{8}$$

where $x_1(t) \in \mathbb{R}^{n_1}$, $x_2(t) \in \mathbb{R}^{n_2}$, $u(t) \in \mathbb{R}^m$, and $0 < \epsilon \ll 1$.

You can write this as a descriptor system:

$$E_1\dot{x} = Ax + Bu \tag{9}$$

where

$$E_1 = \begin{bmatrix} I_{n_1} & 0 \\ 0 & \varepsilon I_{n_2} \end{bmatrix}, \quad A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \tag{10}$$

The input signal is also a linear singularly perturbed system:

$$\begin{aligned} \dot{w}_1 &= S_{11} w_1 + S_{12} w_2 \\ \varepsilon \dot{w}_2 &= S_{21} w_1 + S_{22} w_2 \end{aligned} \tag{11}$$

where $w_1(t) \in \mathbb{R}^{r_1}$, $w_2(t) \in \mathbb{R}^{r_2}$, and $0 < \epsilon \ll 1$. The input to the system is given by $u(t) = L_1 w_1(t) + L_2 w_2(t)$.

The input signal can also be written as a descriptor system like so:

$$E_2 \dot{w} = S w \tag{12}$$

where

$$E_2 = \begin{bmatrix} I_{r_1} & 0 \\ 0 & \varepsilon I_{r_2} \end{bmatrix}, \quad S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}, \quad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}. \tag{13}$$

The output of the system is given by

$$y(t) = C_1 x_1(t) + C_2 x_2(t). \tag{14}$$

3 Wong Sequences

We define the Wong Sequences such:

$$\mathcal{V}_0 = \mathbb{R}^n \qquad \mathcal{V}_{i+1} = A^{-1}(E\mathcal{V}_i) \qquad \mathcal{V}^* = \bigcap_{i=0}^{n} \mathcal{V}_i \tag{15}$$

$$\mathcal{W}_0 = \{0\} \qquad \mathcal{W}_{i+1} = E^{-1}(A\mathcal{W}_i) \qquad \mathcal{W}^* = \bigcup_{i=0}^{n} \mathcal{W}_i \tag{16}$$

4 Wong Sequences: Regular

We now assume that the pencils (E_1, A) and (E_2, S) are regular. Then as shown in the summary by Stephan Trenn, Theorem 2.12:

$$P(sE_1 - A)Q = s \begin{bmatrix} I & 0 \\ 0 & N_1 \end{bmatrix} - \begin{bmatrix} J_1 & 0 \\ 0 & I \end{bmatrix}. \tag{17}$$

$$W(sE_2 - S)T = s \begin{bmatrix} I & 0 \\ 0 & N_2 \end{bmatrix} - \begin{bmatrix} J_2 & 0 \\ 0 & I \end{bmatrix}. \tag{18}$$

The following quantities can then be defined for both the system and the signal.
The index for system is 1, and signal quantities have index 2.
The consistency projector is:

$$\Xi_1 = Q \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \tag{19}$$

The differential selector:

$$\Theta_1 = Q \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} P \tag{20}$$

The impulse selector:

$$\Phi_1 = Q \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} P \tag{21}$$

Furthermore:

$$\tilde{A} := \Theta_1 A \tag{22}$$

$$\hat{E} := \Phi_1 E_1 \tag{23}$$

There are some conditions on (P,Q) related to the Wong sequences. The explicit solution can then be found as the following: Solution for the signal:

$$w(t) = e^{\tilde{S}t}\Xi_2 c_2 \tag{24}$$

The input is then given by:

$$f(t) = BLw(t) = BLe^{\tilde{S}t}\Xi_2 c_2 \tag{25}$$

$$x(t) = e^{\tilde{A}t}\Xi_1 c + \int_0^t e^{\tilde{A}(t-\tau)}\Xi_1 BLe^{\tilde{S}\tau}\Xi_2 c_2 d\tau - \sum_{i=0}^{n-1}\hat{E}^i\Phi_1\left(BLe^{\tilde{S}t}\Xi_2 c_2\right)^{(i)} \tag{26}$$

There are some conditions on c_1, c_2 relating the initial conditions.
We now investigate the integral:

$$\int_0^t e^{\tilde{A}(t-\tau)}\Xi_1 BLe^{\tilde{S}\tau}\Xi_2 c_2 d\tau = e^{\tilde{A}t}\left(\int_0^t e^{-\tilde{A}\tau}\Xi_1 BLe^{\tilde{S}\tau}d\tau\right)\Xi_2 c_2 \tag{27}$$

If $\sigma(\tilde{A}) \cup \sigma\tilde{S} = \emptyset$:

$$\tilde{A}\Pi + \Xi_1 BL = \Pi\tilde{S} \tag{28}$$

Analagous Sylvester Equation?
Then the integral evaluates to:

$$\int_0^t e^{\tilde{A}(t-\tau)}\Xi_1 BLe^{\tilde{S}\tau}\Xi_2 c_2 d\tau = \left(\Pi e^{\tilde{S}t} - e^{\tilde{A}t}\Pi\right)\Xi_2 c_2 \tag{29}$$

This is analagous to the result shown in section 1.
The Sylvester Equation features $\tilde{A} = \Theta_1 A$, $\tilde{S} = \Theta_1 S$. Θ_i takes the upper left partition of A and S. Does this correspond to $(A_{11}, S_{11}, \Pi_{11})$.

5 Concatenated System

For a regular linear system, we can stack the input signal state $w(t)$ and the system state $x(t)$ to create the following autnomous system:

$$\begin{bmatrix}\dot{x}(t)\\ \dot{w}(t)\end{bmatrix} = \begin{bmatrix}A & BL\\ 0 & S\end{bmatrix}\begin{bmatrix}x(t)\\ w(t)\end{bmatrix} \tag{30}$$

.

The explicit solution to this system would then be:

$$\begin{bmatrix}x(t)\\ w(t)\end{bmatrix} = \exp\left(\begin{bmatrix}A & BL\\ 0 & S\end{bmatrix}t\right)\begin{bmatrix}x(0)\\ w(0)\end{bmatrix} \tag{31}$$

Considering the matrix exponential term we have:

$$\exp\left(\begin{bmatrix}A & BL\\ 0 & S\end{bmatrix}t\right) = \begin{bmatrix}e^{At} & \Pi e^{St} - e^{At}\Pi\\ 0 & e^{St}\end{bmatrix} \tag{32}$$

For our descriptor system we then have:

$$\begin{bmatrix}E_1 & 0\\ 0 & E_2\end{bmatrix}\begin{bmatrix}\dot{x}(t)\\ \dot{w}(t)\end{bmatrix} = \begin{bmatrix}A & BL\\ 0 & S\end{bmatrix}\begin{bmatrix}x(t)\\ w(t)\end{bmatrix} \tag{33}$$

Now we reformulate our problem into an analysis of an autonomous descriptor system, namely:

$$\mathcal{E}\dot{\mathbf{X}}(t) = \mathcal{A}\mathbf{X}(t) \tag{34}$$

This should(?) be regular also if (E_1, A) and (E_2, S) is regular. We can find the corresponding quantities, specifically the differential selector for the system defined by $(\mathcal{E}, \mathcal{A})$. By the Stephan Trenn paper we can think of (34) as:

$$\dot{\mathbf{X}}(t) = \Theta\mathcal{A}\mathbf{X}(t) \tag{35}$$

Then it's a matter of comparing $\Theta\mathcal{A}$ and $\begin{bmatrix}A & BL\\ 0 & S\end{bmatrix}$.

So to compare it it would require: computing the wong sequences and finding what Θ will be. Then it's a matter of comparing the two matrices, and exploring how it relates to the Π quantity in (5).

How to compute it?

By the structure of Θ, what terms survive?

$$\Theta = \begin{bmatrix} \Theta_{11} & * \\ 0 & \Theta_{22} \end{bmatrix} \tag{36}$$

Corresponding Θ to $x(t)$ and $w(t)$, and the $*$ term representing the coupling between them.

Relation to $\begin{bmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{21} & \Pi_{22} \end{bmatrix}$?

6 Relation to Scarciotti Paper

Suppose the pencil (S, E_2) is regular. Then we can write the signal generator as the following:

$$\dot{w} = \Theta_2 S w \tag{37}$$

Then, we can write the system as:

$$\begin{aligned} E_1\dot{x} &= Ax + Bu \\ \dot{w} &= \Theta_2 S w \\ y &= Cx \\ u &= Lw \end{aligned} \tag{38}$$

So, we can write the corresponding analagous Sylvester Equation:

$$A\Pi + BL = E_1 \Pi \Theta_2 S \tag{39}$$

and your forward moment becomes:

$$M_g = C\Pi \tag{40}$$

From here, the results follow from Scarciotti's paper.

6.1 When $S_{11} = S_{22} = 0$

The pencil (E_2, S) will be regular in this case when:

$$\det(\lambda E_2 - S) \neq 0 \tag{41}$$

for some λ. Ie:

$$\det\left(\begin{bmatrix} \lambda I & -S_{12} \\ -S_{21} & 0 \end{bmatrix}\right) \neq 0 \tag{42}$$

Using Schur's Complement formula:

$$\begin{aligned} \det \begin{bmatrix} \lambda I & -S_{12} \\ -S_{21} & 0 \end{bmatrix} &= \det(\lambda I) \det(0 - (-S_{21})(\lambda I)^{-1}(-S_{12})) \\ &= \lambda^n \det(-\frac{1}{\lambda} S_{21} S_{12}) \end{aligned} \tag{43}$$

Thus, regularity requires $S_{21}S_{12}$ to be invertible, and both n x n matrices. Which means both S_{12}, S_{21} need to be invertible.

This means that the system:

$$\begin{bmatrix} \dot{w}_1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & S_{12} \\ S_{21} & 0 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \tag{44}$$

will have no dynamics.

$$0 = S_{21} w_1 \implies w_1(t) = 0 \implies w_2(t) = 0 \tag{45}$$

This is all when considering the explicit case $\varepsilon = 0$.